Math for CS & AI: Homework 8

Instructor: Kaifeng Lyu (IIIS, Tsinghua University)

(Adapted from earlier versions of this course taught by Andrew Yao)

Due Date: May 10, 2026

Study Tips

In the lectures on complex analysis, I intentionally skipped some technical details and focused on the high-level ideas. To fill in the missing pieces, you should read the lecture notes carefully.

Some students have asked for additional exercises to practice their problem-solving skills. These exercises are not required, but if you are interested, you may find some useful problems in the following textbooks:

- *Concrete Mathematics: A Foundation for Computer Science*, by Ronald Graham, Donald Knuth, and Oren Patashnik
  - Section 5.4: Generating Functions
  - Chapter 7: Generating Functions
- *An Introduction to Complex Analysis*, by Ravi P. Agarwal, Kanishka Perera, and Sandra Pinelas.
  - Lecture 31: Cauchy's Residue Theorem
  - Lecture 32: Evaluation of Real Integrals by Contour Integration I
  - Lecture 33: Evaluation of Real Integrals by Contour Integration II

Note: Most of these problems are much harder than what is required for this course.

Problem Set

- Exercises 4.11–4.17 from Chapter 4 of the lecture notes:
  https://cloud.tsinghua.edu.cn/d/57ddec714cf74c1cab85/

Writing Guidelines

- Same as the first homework.
- We may deduct up to 3 points for poor writing.
- **Special Note:** Your submission should contain only a single PDF file. Please do not submit other files such as `*.zip` or `*.tex`.

Additional Hints

- We provide additional hints for some of the homework problems in the next few pages.
- If you get stuck on a problem, scroll down to find that problem and read the hints.
- Scroll *very slowly* so that you do not accidentally see hints for problems you have not thought about yet.
- Homework is meant to help you practice your problem-solving skills. It is your job to make sure you have thought enough before reading the hints. Ultimately, there will be no additional hints and no AI tools in the exams.
- Again, **peer learning is strongly encouraged**. Reading hints or getting help from AIs is not the same as learning from your classmate who has thought about the problem and can explain it to you. Make some friends who are willing to discuss problems or even have hotpot with you. Learn from their way of thinking, including both successful and unsuccessful attempts.

The rest of this document contains hints for the homework problems.



Think twice before scrolling down!

Hints

Problem 4.11

- Use Cauchy's integral formula for derivatives:

$$g_n = \frac{1}{2\pi i} \int_\gamma \frac{g(w)}{w^{n+1}} \, \mathrm{d}w = \frac{1}{2\pi i} \int_a^b \frac{g(\gamma(t))}{\gamma(t)^{n+1}} \gamma'(t) \, \mathrm{d}t.$$

- Let $\tilde{\gamma}(t) = g(\gamma(t))$. Can you write the integral as an integral along $\tilde{\gamma}$?
- If you still get stuck, try integrating by parts.

Problem 4.12

- For (a), let $b_{n,k}$ be the number of rooted trees with n nodes such that the root has k children.
- Derive a recurrence relation for $b_{n,k}$, and then the generating function for $b_{n,k}$.
- Add up the generating functions for all k to get the generating function for a_n.

Problem 4.13

- If you do not know how to proceed, read the lecture notes on PGFs!

Problem 4.14

- If you do not know how to proceed, read the lecture notes on Cauchy's Residue Theorem!
- Sometimes you need a semicircle contour, sometimes you need a rectangle contour.

Problem 4.15

- The answer is a trigonometric function.
- It is similar to the generating function of Bernoulli numbers.

Problem 4.16

- A detailed proof sketch is already given in the lecture notes.

Problem 4.17

- A detailed proof sketch is already given in the lecture notes.
